

THE THERMODYNAMICS OF CONTROL IN MULTI-AGENT LANGUAGE MODELS

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Paper under double-blind review

1 METHODOLOGY

Status. This document specifies the next version of the HiddenBench imitation game. It is a *new game derived from*, but not a silent modification of, the currently implemented event-level `hidden_bench_imitation` game. The purpose of the redesign is to place the controller on a slower and more realistic timescale than the microscopic population dynamics while preserving the one-focal-agent imitation geometry and the matched reasoning/classical arms.

1.1 SCIENTIFIC OBJECT

A HiddenBench task provides a scenario, shared information I_s , distributed hidden information I_u , a finite answer set

$$\mathcal{A} = \{a_1, \dots, a_K\},$$

and a correct option o^* . Agent i carries its assigned evidence and a current opinion

$$X_i \in \mathcal{A}.$$

The labelled population state is

$$\mathbf{X} = (X_1, \dots, X_N),$$

and the mesoscopic occupation state is

$$\mathbf{n} = (n_1, \dots, n_K), \quad n_j = \sum_{i=1}^N \mathbf{1}[X_i = a_j], \quad \sum_j n_j = N.$$

As in the existing imitation game, one microscopic update changes at most one focal opinion.

The experiment has two matched dynamical arms:

- reasoning ON: the focal LLM updates from its evidence, history, and current social input,
- reasoning OFF: an explicit provider-free stochastic transition kernel updates the focal opinion.

The task, initialization, population, social scheduler, controller, and recorded observables should otherwise be matched.

1.2 TWO CLOCKS

The redesign introduces an explicit separation between a fast population clock and a slow feedback clock.

Let $k = 0, 1, \dots$ index *population rounds*. Each round contains exactly N microscopic focal-update opportunities, indexed by $r = 1, \dots, N$. Write

$$\mathbf{n}_{k,0} \equiv \mathbf{n}_k, \quad \mathbf{n}_{k,N} \equiv \mathbf{n}_{k+1}.$$

The controller senses and chooses an action only at the round boundary. It does not re-measure or reconsider its decision after individual opinion changes inside that round.

The three principal interaction resources are

$$\begin{aligned} q &= \text{social influence slots per microscopic update,} \\ q_c &= \text{agents sensed once per round,} \end{aligned}$$

and

b = controlled microscopic positions in an ADVOCATE round.

For cross-population comparisons define the intensive sensing and actuation fractions

$$r_s = \frac{q_c}{N}, \quad r_a = \frac{b}{N}.$$

These are distinct resources: q_c controls observation capacity, whereas b controls actuation capacity.

1.3 ROUND-LEVEL FEEDBACK PROTOCOL

At the beginning of round k the controller sees only a noisy population measurement. For a fixed target $Z \in \mathcal{A}$, it samples a set S_k of q_c agents uniformly without replacement and observes

$$Y_k = \{X_i : i \in S_k\}.$$

For target support it is convenient to use

$$s_k = \frac{1}{q_c} \sum_{i \in S_k} \mathbf{1}[X_i = Z].$$

Conditioned on the round-boundary population state, the target-support count is hypergeometric.

The default feedback policy remains stochastic and soft:

$$\Pr(U_k = \text{ADVOCATE}_Z \mid Y_k) = \sigma(\beta(\vartheta - s_k)), \quad \sigma(x) = \frac{1}{1 + e^{-x}},$$

with

$$U_k \in \{\text{NO_OP}, \text{ADVOCATE}_Z\}.$$

The soft policy is scientifically useful because comparable states retain positive support for both actions.

If $U_k = \text{NO_OP}$, all N microscopic updates are ordinary social updates.

If $U_k = \text{ADVOCATE}_Z$, choose exactly

$$B_k \subset \{1, \dots, N\}, \quad |B_k| = b,$$

uniformly at random *once, before the first microscopic update*. The set B_k is the complete actuation schedule for that round. There is no further controller sensing or decision until $k + 1$.

Define

$$C_{k,r} = \mathbf{1}[r \in B_k].$$

At each microscopic position r , sample one focal agent and q distinct ordinary peers from the *current* population state. If $C_{k,r} = 0$, the focal receives all q ordinary social inputs. If $C_{k,r} = 1$, one of the q social slots is replaced by a controller advocacy message for Z , so the focal still receives exactly q influence slots:

$$q = (q - 1) \text{ ordinary peers} + 1 \text{ controller advocacy}.$$

The controller does not overwrite the focal vote. It only changes the local information/social context.

This is a sampled-data, partially observed feedback controller with a fixed per-round intervention budget. The plant evolves on the microscopic clock; measurement and policy updates occur on the slower round clock.

1.4 CONTROLLER SEMANTICS

The controller is external to the population:

- it has no vote;
- it receives no hidden HiddenBench evidence;
- it does not access private agent memories;

- its advocacy message should remain peer-style and should not announce that it is an external controller;
- it never forces the focal answer.

The initial target may be $Z = o^*$. A later wrong-target condition $Z \neq o^*$ should be supported so that control success and truth alignment can be separated.

Microscopic reversibility is *not* part of the game definition. A reversible classical special case may be useful for comparison with the stochastic-thermodynamic framework of Irisarri et al. Irisarri et al. (2026), but the control mechanism should be chosen for plausibility and tractability rather than engineered solely to preserve detailed balance.

1.5 MACROSCOPIC OBSERVABLES

For option Z define

$$p_Z = \frac{n_Z}{N}, \quad m_Z = \frac{Kp_Z - 1}{K - 1}.$$

The principal observables remain

$$m_{\text{truth}} = m_{o^*}, \quad m_{\text{ctrl}} = m_Z,$$

$$m_{\text{order}} = \frac{K \max_j p_j - 1}{K - 1},$$

and normalized vote entropy

$$H_{\text{vote}} = -\frac{\sum_j p_j \ln p_j}{\ln K}.$$

When $Z = o^*$, $m_{\text{ctrl}} = m_{\text{truth}}$; the fields should nevertheless remain distinct to support wrong-target experiments.

1.6 INFORMATION ANALYSIS AT THE CORRECT TIMESCALE

The round-level causal tuple is

$$\boxed{\mathbf{n}_k \longrightarrow Y_k \longrightarrow U_k \longrightarrow \mathbf{n}_{k+1}.$$

The primary controller-to-population information quantity is therefore

$$\boxed{\mathcal{T}_{U \rightarrow N}^{\text{round}} = I(U_k; \mathbf{n}_{k+1} \mid \mathbf{n}_k)}$$

in bits per population round. In a Markov classical model this is the one-step transfer entropy at the feedback timescale. In the reasoning arm it is the corresponding first-order empirical directed-information statistic; a later history-conditioned analysis can test whether the round-boundary state is insufficient because of LLM memory.

The corresponding round-level sensing quantity is

$$I(\mathbf{n}_k; Y_k).$$

Useful lower-dimensional actuation projections are

$$I(U_k; n_{Z,k+1} \mid n_{Z,k}), \quad I(U_k; n_{\text{truth},k+1} \mid n_{\text{truth},k}),$$

together with

$$H(U_k \mid \mathbf{n}_k)$$

and the information fraction

$$\rho_N = \frac{I(U_k; \mathbf{n}_{k+1} \mid \mathbf{n}_k)}{H(U_k \mid \mathbf{n}_k)}.$$

The ratio is a normalization diagnostic, not a thermodynamic efficiency.

A signed round response should accompany the unsigned CMI, for example the state-adjusted difference

$$\mathbb{E}[\Delta m_Z \mid U_k = \text{ADVOCATE}_Z, \mathbf{n}_k] - \mathbb{E}[\Delta m_Z \mid U_k = \text{NO_OP}, \mathbf{n}_k]$$

over states with action overlap.

Why this is a new analysis layer. The old event-level statistic $I(U_t; \mathbf{n}_{t+1} \mid \mathbf{n}_t)$ cannot simply be averaged or summed to obtain the round quantity. All N microscopic transitions in a controlled round share one action and are coupled through the evolving state. Therefore the new game must persist one explicit round record and compute the information statistics from round-boundary samples.

1.7 MICROSCOPIC AND EPISODE DIAGNOSTICS

The microscopic event log should be retained. It is useful for asking what the preallocated intervention actually does locally. Within ADVOCATE rounds, $C_{k,r}$ is randomized in advance, so a useful local diagnostic is

$$I(C_{k,r}; X_{k,r+1}^f \mid X_{k,r}^f, \mathbf{n}_{k,r}, U_k = \text{ADVOCATE}_Z),$$

plus signed focal target-adoption/target-leaving responses. This is a local intervention diagnostic, not the headline round-level transfer entropy.

Episode-level trajectory quantities should also remain, including final truth alignment and the net truth current

$$J_{\text{truth}} = N_{\rightarrow \text{truth}} - N_{\leftarrow \text{truth}},$$

together with activity

$$A_{\text{truth}} = N_{\rightarrow \text{truth}} + N_{\leftarrow \text{truth}}.$$

Because J_{truth} telescopes under one-focal updates, activity is required to distinguish quiet and highly fluctuating paths with the same endpoint.

1.8 ESTIMATION AND SUPPORT

Round-level information estimates should be pooled across repeated episodes within a fixed experimental cell, while uncertainty is obtained by resampling whole episodes. They should not be estimated separately from one short episode.

The coarser feedback clock reduces the number of information samples by roughly a factor of N relative to the old event-level analysis. Consequently:

- full-state CMI may be sparse for large N and K ;
- target-count projections are important robustness diagnostics;
- action-overlap and singleton/sparsity diagnostics must be reported;
- more repeated episodes may be required than in the event-level game.

Because the stochastic policy is known, a natural null for actuation is a conditional-randomization null that resamples

$$U_k^* \sim \pi(\cdot \mid Y_k)$$

from the logged policy probability while holding the observed state and outcome records fixed. This preserves the measurement-to-policy coupling while destroying the observed action-to-outcome association. Episode-level bootstrap uncertainty should remain the default uncertainty calculation.

1.9 SCIENTIFIC COMPARISON

The central controlled comparison is

	no feedback	round feedback
reasoning ON	A	B
reasoning OFF	C	D

with matched task, initialization, N , q , round horizon, and seeds where possible. The main structural control axes are now

$$(q, q_c/N, b/N),$$

with N providing the finite-size dimension.

The theory target is an explicit classical round kernel

$$R_u(\mathbf{n}_{k+1} \mid \mathbf{n}_k)$$

and its exact or approximate transfer entropy. The LLM experiment then asks whether semantic reasoning changes controllability, fluctuations, and the adequacy of the Markov round-boundary description.

1.10 WHAT THIS REDESIGN DOES AND DOES NOT CLAIM

The redesign is motivated by a clearer physical interpretation of feedback: a controller observes the population, commits a finite intervention budget for a slower interval, lets the population evolve, and measures again. It also separates sensing capacity from actuation capacity, which were not independent resources in the old event-level protocol.

It does *not* invalidate the existing event-level results. Those results belong to a different controller/game and should remain reproducible under the old `hidden_bench_imitation` type. Numbers from the old event-level CMI and the new round-level CMI should not be presented as the same estimator at a different plotting resolution.

REFERENCES

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A THEORY

Purpose of this note. We now fix the classical response rule instead of leaving it abstract. Ordinary agents follow a unanimity q -voter rule: a focal agent switches opinion only when all q social inputs unanimously support the opposite opinion. At a controlled update, the controller does not change this rule; it replaces one of the q social inputs by advocacy for the target opinion. The controller senses and decides only once per population round, and, if it chooses ADVOCATE, exactly b of the N microscopic update positions are selected uniformly in advance. The goal is to derive the one-step kernels K_0, K_1 , the full-round kernels R_0, R_1 , and the controller-to-population transfer entropy. Microscopic reversibility is *not* imposed.

A.1 MODEL AND NOTATION: WHAT EVOLVES, AND ON WHICH TIME SCALE?

In simple terms. There are two time scales. Inside a round, agents update one at a time. The controller, however, acts on a slower time scale: it measures once at the beginning of the round, decides whether to advocate the target opinion, and then lets the whole round unfold without reconsidering its decision.

We first derive the binary model. Let opinion A be the controller’s target and B the alternative. There are N agents. At the beginning of round k ,

$$n_k = n \in \{0, 1, \dots, N\}$$

means that exactly n agents currently support A and $N - n$ support B . Each round contains exactly N microscopic update attempts, indexed by $r = 0, \dots, N - 1$. Thus capital N is the fixed population size, while lower-case n_k is the stochastic population state observed on the round clock.

The main variables are:

Symbol	Meaning
N	Population size; also the number of microscopic update positions in one round.
q	Number of social inputs seen by the focal agent at one microscopic update.
q_c	Number of agents sampled by the controller at the beginning of a round.
b	Exact number of controlled microscopic positions if the controller chooses ADVOCATE.
$c = b/N$	Actuation fraction: controlled positions per round.
n_k	Number of target supporters at the beginning of round k .
Y_k	Number of target supporters found in the controller’s sample of size q_c .
U_k	Round-level action, $U_k \in \{\text{NOOP}, \text{ADVOCATE}\}$.
K_0	Transition kernel for <i>one ordinary microscopic update</i> .
K_1	Transition kernel for <i>one controlled microscopic update</i> .
R_0, R_1	Transition kernels for a <i>whole round</i> under NOOP or ADVOCATE.

The round protocol is

$$n_k \longrightarrow Y_k \longrightarrow U_k \longrightarrow N \text{ microscopic updates} \longrightarrow n_{k+1}.$$

If $U_k = \text{ADVOCATE}$, a set $S_k \subset \{1, \dots, N\}$ with $|S_k| = b$ is drawn uniformly at random *before* any microscopic update occurs. Positions in S_k use the controlled kernel K_1 ; all other positions use K_0 . No new sensing or controller decision occurs within the round.

How to validate this section. In simulation logs, each round should contain exactly N focal-update attempts. There must be exactly one controller sensing/policy evaluation per round. Under ADVOCATE, exactly b positions should be marked controlled and their indices should be fixed before the first update. Under NOOP, there should be zero controlled positions. These are protocol-level invariants and should hold for every random seed.

A.2 THE FIXED CLASSICAL RULE AND THE MICROSCOPIC KERNELS K_0, K_1

THE UNANIMITY q -VOTER RULE

In simple terms. At every microscopic update, one focal agent looks at q social inputs. It changes opinion only if all q inputs agree with each other and all oppose the focal’s current opinion. The controller does not create a new decision rule; at a controlled position it simply occupies one of those q input slots and says “A”.

This is the classical herding part of a q -voter construction: the probability of switching depends nonlinearly on finding q opposite opinions, as in the social-imitation model of Irisarri et al. Irisarri et al. (2026). We deliberately do *not* add their anticonformity reaction here, because our immediate objective is a minimal classical analogue of the advocacy intervention.

For an ordinary position:

1. choose one focal agent uniformly from the N agents;
2. sample q distinct ordinary peers uniformly from the remaining $N - 1$ agents;
3. switch the focal only if all q peers unanimously hold the opposite opinion.

For a controlled position with target A :

1. choose the focal in exactly the same way;
2. one of the q input slots is occupied by the controller and therefore supports A ;
3. sample only $q - 1$ ordinary peers from the remaining $N - 1$ agents;
4. apply the *same* unanimity rule to the resulting q inputs.

ORDINARY MICROSCOPIC KERNEL K_0

What we calculate now. Starting from n supporters of A , one microscopic update can only produce $n + 1$, n , or $n - 1$. We calculate these three probabilities exactly.

A transition $n \rightarrow n + 1$ requires (i) choosing a B focal and (ii) sampling q peers that are all A . The probability of choosing a B focal is $(N - n)/N$. Conditional on that choice, all n supporters of A

remain available among the other $N - 1$ agents. Therefore

$$K_0(n + 1 | n) = \frac{N - n}{N} \frac{\binom{n}{q}}{\binom{N-1}{q}} \equiv p_0^+(n). \quad (1)$$

Similarly, $n \rightarrow n - 1$ requires an A focal followed by q unanimously B peers. Hence

$$K_0(n - 1 | n) = \frac{n}{N} \frac{\binom{N-n}{q}}{\binom{N-1}{q}} \equiv p_0^-(n). \quad (2)$$

The remaining probability is a self-transition,

$$K_0(n | n) = 1 - p_0^+(n) - p_0^-(n). \quad (3)$$

We use the convention $\binom{a}{q} = 0$ when $a < q$.

CONTROLLED MICROSCOPIC KERNEL K_1

What changes under control. The controller contributes one guaranteed A input. A B focal therefore needs only the remaining $q - 1$ ordinary peers to support A in order to see q unanimous A inputs. An A focal can no longer see q unanimous B inputs, because one input is fixed to A by the controller.

Thus

$$K_1(n + 1 | n) = \frac{N - n}{N} \frac{\binom{n}{q-1}}{\binom{N-1}{q-1}} \equiv p_1^+(n), \quad (4)$$

while

$$K_1(n - 1 | n) = 0, \quad (5)$$

and

$$K_1(n | n) = 1 - p_1^+(n). \quad (6)$$

This controlled kernel is asymmetric and may be microscopically irreversible. That is acceptable: transfer entropy only requires well-defined stochastic transition probabilities. Irisarri et al.'s reversibility structure is needed for their specific affinity/fluctuation-theorem construction, not for Eqs. equation 4–equation 6 themselves Irisarri et al. (2026).

A concrete example. Take $N = 6$, $q = 2$, and $n = 2$. In an ordinary update,

$$p_0^+(2) = \frac{4}{6} \frac{\binom{2}{2}}{\binom{5}{2}} = \frac{1}{15}, \quad p_0^-(2) = \frac{2}{6} \frac{\binom{4}{2}}{\binom{5}{2}} = \frac{1}{5}.$$

So when A is a minority, the ordinary q -voter tends to reduce it further. At a controlled position,

$$p_1^+(2) = \frac{4}{6} \frac{\binom{2}{1}}{\binom{5}{1}} = \frac{4}{15}, \quad p_1^-(2) = 0.$$

The controller therefore changes the local drift without changing the focal response rule.

How to validate K_0 and K_1 . First check analytically that every row is nonnegative and sums to one. Then, for fixed (N, q, n) , simulate 10^5 isolated microscopic updates and compare empirical frequencies of $n + 1, n, n - 1$ against Eqs. equation 1–equation 6. Useful hand checks are: (i) $q = 1$ gives $p_0^+ = p_0^- = n(N - n)/[N(N - 1)]$ and $p_1^+ = (N - n)/N$; (ii) at $n = N$, both kernels remain at N ; (iii) for $q > 1$ and $n = 0$, even control cannot create the first A supporter because $q - 1$ ordinary A peers are still required.

A.3 FROM MICROSCOPIC KERNELS TO WHOLE-ROUND KERNELS R_0, R_1

In simple terms. K_0 and K_1 describe only one focal update. Transfer entropy is defined on our slower round clock, so we need the probability of ending the entire round at m supporters, given that it started at n supporters. This is what $R_0(m | n)$ and $R_1(m | n)$ mean.

Represent K_0 and K_1 as $(N + 1) \times (N + 1)$ transition matrices with rows indexed by the current n and columns by the next m .

NO-CONTROL ROUND

If $U_k = \text{NOOP}$, all N microscopic positions are ordinary. Therefore the exact round kernel is

$$R_0 = K_0^N, \quad R_0(m | n) = [K_0^N]_{nm}. \quad (7)$$

ADVOCACY ROUND WITH EXACTLY b CONTROLLED POSITIONS

If $U_k = \text{ADVOCATE}$, exactly b positions use K_1 and $N - b$ use K_0 . Because the population changes after every update, the order of K_0 and K_1 generally matters. We therefore average over the uniformly preallocated schedule exactly.

Let $F_{r,j}$ be an $(N + 1) \times (N + 1)$ matrix whose (n, m) element is the probability of being at m after r microscopic positions, having used exactly j controlled positions, conditional on starting the round at n . Initialize

$$F_{0,0} = I,$$

where I is the identity matrix. After r positions and j controlled uses, there are $b - j$ controlled positions and $(N - b) - (r - j)$ ordinary positions remaining. Hence

$$F_{r+1,j+1} += \frac{b-j}{N-r} F_{r,j} K_1, \quad (8)$$

$$F_{r+1,j} += \frac{N-b-r+j}{N-r} F_{r,j} K_0. \quad (9)$$

After all N positions,

$$R_1 = F_{N,b}. \quad (10)$$

This recursion is exactly equivalent to averaging over all $\binom{N}{b}$ possible schedules, but it avoids enumerating them.

Two limiting identities are useful:

$$b = 0 \Rightarrow R_1 = R_0, \quad b = N \Rightarrow R_1 = K_1^N.$$

How to validate R_0 and R_1 . For tiny systems such as $N \leq 6$, enumerate all $\binom{N}{b}$ schedules explicitly and verify that their averaged ordered matrix products equal the recursion in Eq. equation 10. For realistic N , fix an initial n , simulate many complete rounds, and compare the histogram of n_{k+1} with the corresponding row of R_0 or R_1 . Also verify the identities $b = 0 \Rightarrow R_1 = R_0$ and $b = N \Rightarrow R_1 = K_1^N$ numerically to machine precision.

A.4 CONTROLLER SENSING AND THE CLOSED-LOOP ROUND PROCESS

In simple terms. Before choosing NOOP or ADVOCATE, the controller sees only a sample of the population. We first calculate how likely each sample result is, then average the controller's policy over that measurement noise. This gives the action probability $p(u | n)$ needed by the transfer-entropy calculation.

At the beginning of the round, the controller samples q_c distinct agents without replacement. If $Y_k = y$ of them support A , then

$$S_{q_c}(y | n) = \mathbb{P}(Y_k = y | n_k = n) = \frac{\binom{n}{y} \binom{N-n}{q_c-y}}{\binom{N}{q_c}}. \quad (11)$$

This is the hypergeometric sensor model.

Let $\pi_1(y) = \mathbb{P}(U_k = \text{ADVOCATE} \mid Y_k = y)$ denote the controller policy. We do not need to fix a particular form for the theory. A useful soft-threshold example is

$$\pi_1(y) = \sigma \left[\eta \left(\vartheta - \frac{y}{q_c} \right) \right], \quad \sigma(z) = \frac{1}{1 + e^{-z}},$$

where ϑ is a target-support threshold and η controls policy sharpness.

The probability of advocacy conditioned on the *true* population state is

$$p_1(n) \equiv \mathbb{P}(U_k = \text{ADVOCATE} \mid n_k = n) = \sum_y S_{q_c}(y \mid n) \pi_1(y). \quad (12)$$

The action-averaged closed-loop transition kernel is therefore row-wise

$$R(m \mid n) = [1 - p_1(n)] R_0(m \mid n) + p_1(n) R_1(m \mid n). \quad (13)$$

If $P_k(n) = \mathbb{P}(n_k = n)$ is the state distribution before round k , then

$$P_{k+1}(m) = \sum_n P_k(n) R(m \mid n). \quad (14)$$

How to validate sensing and policy. Hold the true state n fixed and repeatedly sample q_c agents. The empirical distribution of Y should match Eq. equation 11. Next run the policy on those samples and compare the empirical advocacy frequency with Eq. equation 12. When $q_c = N$, the sensor becomes exact: $Y = n$ with probability one. When q_c is small, repeated samples should visibly broaden the action distribution. This directly checks the intended sensing-resource interpretation of q_c .

A.5 EXACT ROUND-LEVEL TRANSFER ENTROPY

What transfer entropy means here. We ask: after the current population state n_k is already known, how much additional information does knowing the controller's round action U_k provide about the next population state n_{k+1} ? If NOOP and ADVOCATE lead to nearly identical next-state distributions, the answer is small. If they lead to clearly different next-state distributions, the answer is large.

For our round-level Markov description, the controller-to-population transfer-entropy increment is

$$\mathcal{T}_k = I(U_k; n_{k+1} \mid n_k). \quad (15)$$

Using $P_k(n)$, the action probabilities, and the exact round kernels gives the finite sum. The steps are

$$\begin{aligned} I(U_k; n_{k+1} \mid n_k) &= \sum_{n, u, m} \mathbb{P}(n, u, m) \log_2 \frac{\mathbb{P}(m \mid n, u)}{\mathbb{P}(m \mid n)}, \\ \mathbb{P}(n, u, m) &= P_k(n) p(u \mid n) R_u(m \mid n), \\ \mathbb{P}(m \mid n) &= R(m \mid n). \end{aligned}$$

Substituting these three pieces yields

$$\mathcal{T}_k = \sum_{n, u, m} P_k(n) p(u \mid n) R_u(m \mid n) \log_2 \frac{R_u(m \mid n)}{R(m \mid n)}. \quad (16)$$

This is an *exact finite- N expression* in bits per round once K_0, K_1 , the sensor, and the policy are fixed. Equivalently,

$$\mathcal{T}_k = \sum_n P_k(n) \sum_u p(u \mid n) D_{\text{KL}}^{(2)}(R_u(\cdot \mid n) \parallel R(\cdot \mid n)), \quad (17)$$

where $D_{\text{KL}}^{(2)}$ uses logarithm base two.

Equation equation 17 gives the most useful interpretation: **transfer entropy is the average distinguishability between the transition law selected by the controller and the transition law obtained when the controller action is not revealed.**

For an experiment of K rounds, define the finite-horizon rate

$$\overline{\mathcal{T}}_{U \rightarrow N}^{(K)} = \frac{1}{K} \sum_{k=0}^{K-1} \mathcal{T}_k. \quad (18)$$

This finite-horizon quantity is the natural headline measure for the pure unanimity q -voter baseline.

Why finite-horizon TE is important here. A pure q -voter has absorbing consensus states. With persistent advocacy for A , $n = N$ is absorbing, and the long-time stationary distribution may collapse onto consensus. Once the population is already at $n = N$, $R_0(\cdot | N) = R_1(\cdot | N)$ and the stationary actuation TE is zero. This does *not* mean the controller had no influence on the path to consensus. It means that no new control information is transmitted after the system has become frozen. Therefore we should primarily report $\mathcal{T}_k$ as a function of round and its finite-horizon average in Eq. equation 18. A non-degenerate stationary TE would require adding spontaneous noise/anticonformity or another mechanism that prevents absorption.

The TE obeys

$$0 \leq \mathcal{T}_k \leq H(U_k | n_k). \quad (19)$$

If $R_0 = R_1$ (for example $b = 0$), then $\mathcal{T}_k = 0$. If the action is completely deterministic once n_k is known, then $H(U_k | n_k) = 0$ and therefore $\mathcal{T}_k = 0$ even if the chosen action has a large causal effect. In our feedback design, randomness can remain because the controller observes only the random sample Y_k and/or because the policy itself is stochastic. This distinction is important when interpreting TE as an information-flow measure rather than as a generic causal-effect size.

Horowitz and Sandberg use transfer entropy in feedback thermodynamics mainly for the opposite direction, system $\rightarrow$ measurement/controller, and derive the continuous construction by discretizing repeated feedback Horowitz & Sandberg (2014). Our Eq. equation 16 is the *actuation* direction, controller $\rightarrow$ population. The sensing direction should be reported separately, e.g. through $I(n_k; Y_k)$.

How to validate the transfer entropy. There are three strong checks. (1) Generate long synthetic trajectories directly from the exact closed-loop kernel and estimate $I(U_k; n_{k+1} | n_k)$ by contingency tables; it should converge to Eq. equation 16. (2) Shuffle U_k within appropriate state strata: the estimated TE should collapse toward its finite-sample null. (3) Check exact identities: $b = 0 \Rightarrow \mathcal{T}_k = 0$; $R_0 = R_1 \Rightarrow \mathcal{T}_k = 0$; and every numerical value must satisfy Eq. equation 19. These tests validate both the derivation and the estimator already used in the repository’s direct-counting information-analysis path.

A.6 LARGE- N APPROXIMATION: WHAT PHASE STRUCTURE SHOULD WE EXPECT?

In simple terms. The exact kernels tell us what happens for finite N . To understand phase behavior, we now replace the discrete fraction n/N by a continuous variable $x \in [0, 1]$ and calculate the average drift produced by ordinary and controlled microscopic updates. This is the mean-field layer; it is an approximation to the exact finite-budget model, not a replacement for it.

Let

$$x = \frac{n}{N}.$$

For large N with fixed q , hypergeometric sampling is approximately sampling from proportions. The ordinary microscopic upward and downward probabilities become

$$p_0^+(x) \simeq (1-x)x^q, \quad (20)$$

$$p_0^-(x) \simeq x(1-x)^q. \quad (21)$$

Hence the ordinary drift in the intensive variable is proportional to

$$f_0(x) = (1-x)x^q - x(1-x)^q. \quad (22)$$

For a controlled position,

$$f_1(x) = (1-x)x^{q-1}, \quad (23)$$

because only $B \rightarrow A$ switches remain possible.

During an ADVOCATERound, a fraction $c = b/N$ of positions are controlled. At leading mean-field order the within-round drift is therefore

$$f_{\text{ADVOCATE}}(x; c) = (1 - c)f_0(x) + cf_1(x), \quad (24)$$

whereas $f_{\text{NOOP}}(x) = f_0(x)$. If one unit of rescaled time corresponds to one population round, the mean-field round maps Φ_0 and Φ_1 are obtained by integrating

$$\frac{dx}{d\tau} = f_{\text{NOOP}}(x) \quad \text{or} \quad \frac{dx}{d\tau} = f_{\text{ADVOCATE}}(x; c), \quad \tau \in [0, 1].$$

The controller still chooses the action only once at the beginning of the round. Therefore the next-state law is a mixture of the two round maps, not a policy decision at each infinitesimal time.

A SIMPLE ANALYTIC CONTROLLABILITY THRESHOLD

For $x < 1/2$, the ordinary q -voter drift $f_0(x)$ is negative when $q > 1$: a minority tends to shrink. We can ask how large c must be for an ADVOCATERound to reverse that local drift. Solve $f_{\text{ADVOCATE}}(x; c) > 0$. For $q \geq 2$ and $0 < x < 1/2$ this gives

$$c > c_*(x; q) = \frac{(1 - x)^{q-1} - x^{q-1}}{(1 - x)[x^{q-2} + (1 - x)^{q-2}]}. \quad (25)$$

For $q = 2$,

$$c_*(x; 2) = \frac{1 - 2x}{2(1 - x)},$$

whereas for $q = 3$,

$$c_*(x; 3) = \frac{1 - 2x}{1 - x}.$$

This already predicts that stronger social unanimity makes a small target minority substantially harder to control. It also exposes an important structural fact: for $q > 1$ and $x = 0$, one-slot advocacy cannot create the first target supporter because the remaining $q - 1$ ordinary peers must already include target support. The controller can amplify an existing seed but, except at $q = 1$, cannot create one from nothing.

This suggests the theoretical phase/controllability surfaces to study first:

$$\overline{T}_{U \rightarrow N}^{(K)}(q, q_c, c), \quad \mathbb{E}[x_K](q, q_c, c), \quad \text{Var}(x_K)(q, q_c, c),$$

and, in the mean-field approximation, the location of drift reversals or basin changes as functions of (q, c) with sensing q_c entering through the policy.

How to validate the mean-field layer. For several N values (for example $N = 20, 50, 100, 200$), compare: (i) the exact one-step drifts computed from K_0, K_1 with Eqs. equation 22–equation 23; (ii) Monte Carlo average trajectories with the ODE round maps Φ_0, Φ_1 ; and (iii) the empirical control threshold for positive drift against $c_*(x; q)$. The discrepancy should shrink with increasing N away from absorbing boundaries. This is the cleanest way to justify, rather than merely assume, the mean-field approximation.

A.7 DIRECT EXTENSION TO THE THREE-OPTION HIDDENBENCH SETTING

In simple terms. HiddenBench has three decision options rather than two Li et al. (2026). The binary calculation above is the easiest place to understand the theory, but the fixed unanimity rule extends directly. The population state becomes a count vector instead of a single count.

Let

$$\mathbf{n} = (n_A, n_B, n_C), \quad n_A + n_B + n_C = N,$$

with A the controller target. In an ordinary microscopic update, a focal agent in option i changes to option $j \neq i$ only if all q ordinary peers unanimously support j . Therefore

$$K_0(\mathbf{n} - \mathbf{e}_i + \mathbf{e}_j \mid \mathbf{n}) = \frac{n_i}{N} \frac{\binom{n_j}{q}}{\binom{N-1}{q}}, \quad i \neq j. \quad (26)$$

At a controlled position, one input is fixed to A . The only possible opinion change is therefore from a non- A option i to A :

$$K_1(\mathbf{n} - \mathbf{e}_i + \mathbf{e}_A \mid \mathbf{n}) = \frac{n_i}{N} \frac{\binom{n_A}{q-1}}{\binom{N-1}{q-1}}, \quad i \neq A. \quad (27)$$

All other off-diagonal controlled transitions are zero, and the diagonal element completes each row to one. The same finite- b recursion then constructs $R_1(\mathbf{m} \mid \mathbf{n})$, and Eq. equation 16 remains unchanged after replacing scalar counts by composition vectors.

The number of three-option occupation states is

$$\binom{N+2}{2},$$

so for the population sizes currently under consideration the exact state space is still manageable, especially because K_0 and K_1 are sparse.

How to validate the three-option extension. Use tiny populations and enumerate all focal/peer choices directly. Verify Eq. equation 26 and Eq. equation 27 against empirical microscopic transitions. Then compare exact round kernels with Monte Carlo as in the binary case. Finally, check permutation symmetry between the two non-target options B and C when their initial counts and evidence treatment are symmetric.

A.8 WHAT THE THEORY NOW GIVES US, AND WHAT REMAINS EMPIRICAL

The fixed-rule theory now separates four objects cleanly:

1. **Social dynamics:** q determines how much unanimous peer support is needed for imitation.
2. **Sensing resource:** q_c determines how noisy the controller’s round-level observation is.
3. **Actuation resource:** $c = b/N$ determines how many microscopic opportunities receive advocacy.
4. **Information flow:** $\mathcal{T}_k$ measures how distinguishable the next-round population dynamics are when the controller action is known.

The exact finite- N theory is algorithmic but not necessarily a single elementary closed-form scalar: K_0 and K_1 are explicit, R_0 is a matrix power, R_1 is obtained by an exact finite recursion, and transfer entropy is then the explicit finite sum in Eq. equation 16. This is sufficient to generate theoretical TE surfaces without Monte Carlo estimation error.

For the LLM experiment, the same round-level protocol should be used with reasoning OFF and ON. The classical q -voter then provides a tractable reference. Reasoning is not represented by inserting an arbitrary sigmoid into the classical model; instead, its effect is diagnosed empirically by where the LLM transition statistics, memory dependence, outcome drift, variance, and information flow depart from the classical predictions.

Recommended immediate validation sequence. (1) Implement the binary K_0, K_1 formulas and unit-test them. (2) Build the exact R_0, R_1 recursion and compare it with brute force for small N . (3) Generate exact finite-horizon TE from Eq. equation 16. (4) Simulate the same classical process and verify that empirical CMI converges to the exact TE. (5) Produce (q, q_c, c) controllability surfaces from both exact finite- N theory and mean field. (6) Only then compare reasoning OFF/ON under the identical slow feedback protocol.